

Lecture 19:

Data Visualization (in Python)



INFO 1100: Introduction to Programming

we're down to the final topic for this class!

now that we've learned something about designing
good visualizations...

now that we've learned something about designing
good visualizations...

how do we create them?

now that we've learned something about designing
good visualizations...

how do we create them?

^
in Python

let's learn about

Matplotlib

let's learn about
Matplotlib

most commonly used package for data viz in Python!

Creating figures

Creating figures

import the package:

```
import matplotlib.pyplot as plt
```

Creating figures

there are several ways of creating figures

Creating figures

there are several ways of creating figures

like the reading, we'll look at the **object-oriented
interface**

Creating figures

there are several ways of creating figures

like the reading, we'll look at the **object-oriented
interface**

more complex + more powerful!

Creating figures

```
import matplotlib.pyplot as plt
```

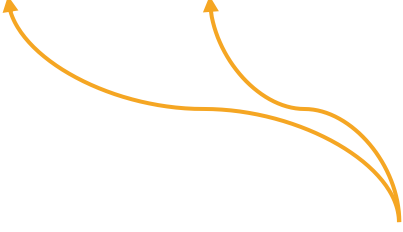
Creating figures

```
import matplotlib.pyplot as plt  
  
fig, ax = plt.subplots()
```

Creating figures

```
import matplotlib.pyplot as plt
```

```
fig, ax = plt.subplots()
```

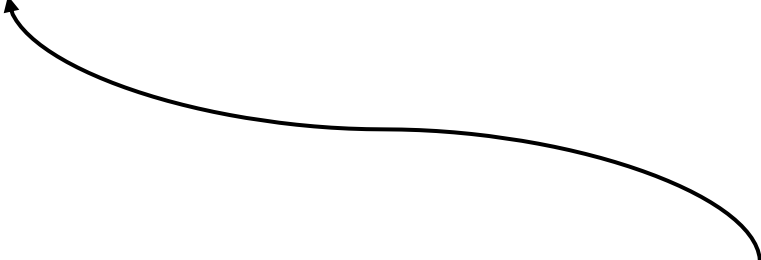


two objects!

Creating figures

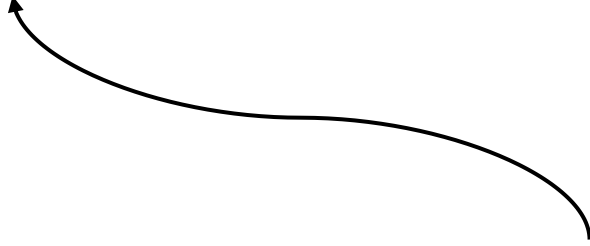
```
import matplotlib.pyplot as plt  
fig, ax = plt.subplots()
```

top-level container
(like the gallery wall)



Creating figures

```
import matplotlib.pyplot as plt  
fig, ax = plt.subplots()
```



individual plot(s)
(the works of art)

Creating figures

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots()

a = [1, 2, 3, 4, 5]
b = [6.2, 5.3, 2.6, 4.8, 5.7]
```

Creating figures

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots()

a = [1, 2, 3, 4, 5]
b = [6.2, 5.3, 2.6, 4.8, 5.7]

ax.plot(a, b)
```

Creating figures

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots()

a = [1, 2, 3, 4, 5]
b = [6.2, 5.3, 2.6, 4.8, 5.7]

ax.plot(a, b)
```

use the **ax** object to plot

Creating figures

```
import matplotlib.pyplot as plt  
  
fig, ax = plt.subplots()  
  
a = [1, 2, 3, 4, 5]  
b = [6.2, 5.3, 2.6, 4.8, 5.7]  
  
ax.plot(a, b)
```

`plot()` creates a line chart

Creating figures

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots()

a = [1, 2, 3, 4, 5]
b = [6.2, 5.3, 2.6, 4.8, 5.7]

ax.plot(a, b)
```

first argument is the x-axis data

Creating figures

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots()

a = [1, 2, 3, 4, 5]
b = [6.2, 5.3, 2.6, 4.8, 5.7]

ax.plot(a, b)
```

second argument is the y-axis data

Creating figures

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots()

a = [1, 2, 3, 4, 5]
b = [6.2, 5.3, 2.6, 4.8, 5.7]

ax.plot(a, b)
plt.show()
```


Creating figures

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots()

a = [1, 2, 3, 4, 5]
b = [6.2, 5.3, 2.6, 4.8, 5.7]

ax.plot(a, b)
plt.show()
```

`show()` displays the chart

Creating figures

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots()

a = [1, 2, 3, 4, 5]
b = [6.2, 5.3, 2.6, 4.8, 5.7]

ax.plot(a, b)
plt.show()
```

method of the class, not
either object!

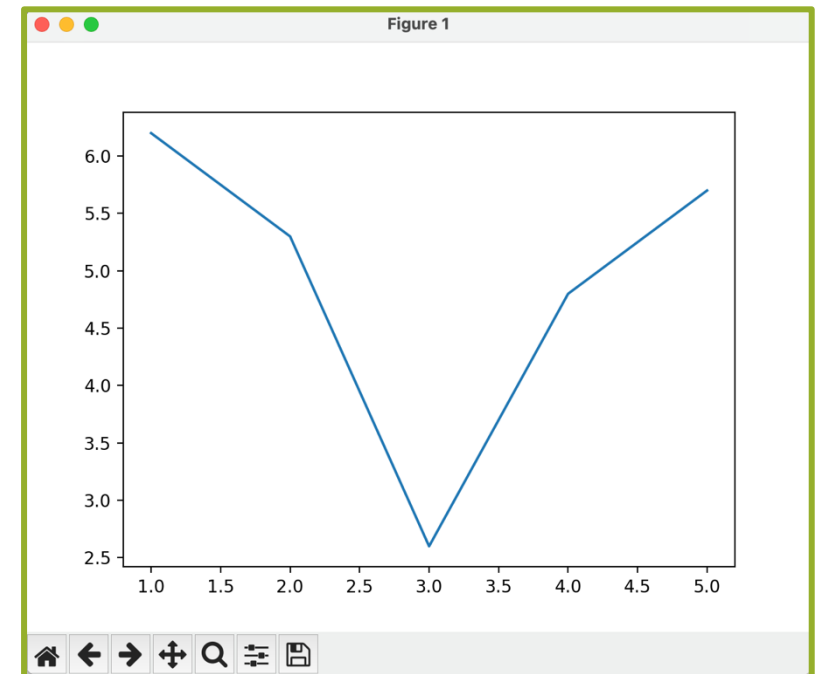
Creating figures

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots()

a = [1, 2, 3, 4, 5]
b = [6.2, 5.3, 2.6, 4.8, 5.7]

ax.plot(a, b)
plt.show()
```



Creating figures

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots()

a = [1, 2, 3, 4, 5]
b = [6.2, 5.3, 2.6, 4.8, 5.7]

ax.plot(a, b)
```

we can also change the size of the figure

Creating figures

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots(figsize=(6,2))

a = [1, 2, 3, 4, 5]
b = [6.2, 5.3, 2.6, 4.8, 5.7]

ax.plot(a, b)
```

Creating figures

```
import matplotlib.pyplot as plt
```

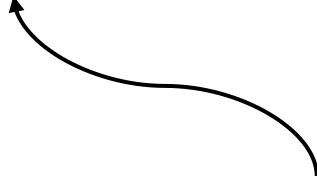
```
fig, ax = plt.subplots(figsize=(6,2))
```

```
a = [1, 2, 3, 4, 5]
```

```
b = [6.2, 5.3, 2.6, 4.8, 5.7]
```

```
ax.plot(a, b)
```

width



Creating figures

```
import matplotlib.pyplot as plt
```

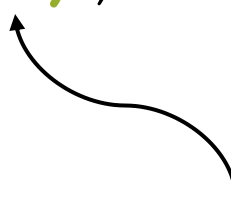
```
fig, ax = plt.subplots(figsize=(6,2))
```

```
a = [1, 2, 3, 4, 5]
```

```
b = [6.2, 5.3, 2.6, 4.8, 5.7]
```

```
ax.plot(a, b)
```

height



Creating figures

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots(figsize=(6,2))

a = [1, 2, 3, 4, 5]
b = [6.2, 5.3, 2.6, 4.8, 5.7]

ax.plot(a, b)
plt.show()
```

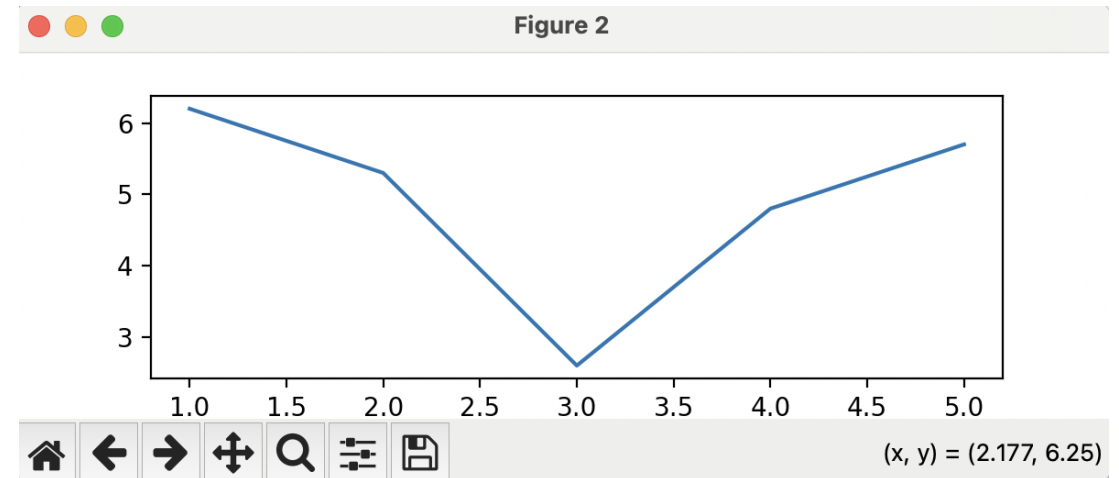

Creating figures

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots(figsize=(6,2))

a = [1, 2, 3, 4, 5]
b = [6.2, 5.3, 2.6, 4.8, 5.7]

ax.plot(a, b)
plt.show()
```



Creating figures

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots(figsize=(4, 6))

a = [1, 2, 3, 4, 5]
b = [6.2, 5.3, 2.6, 4.8, 5.7]

ax.plot(a, b)
plt.show()
```

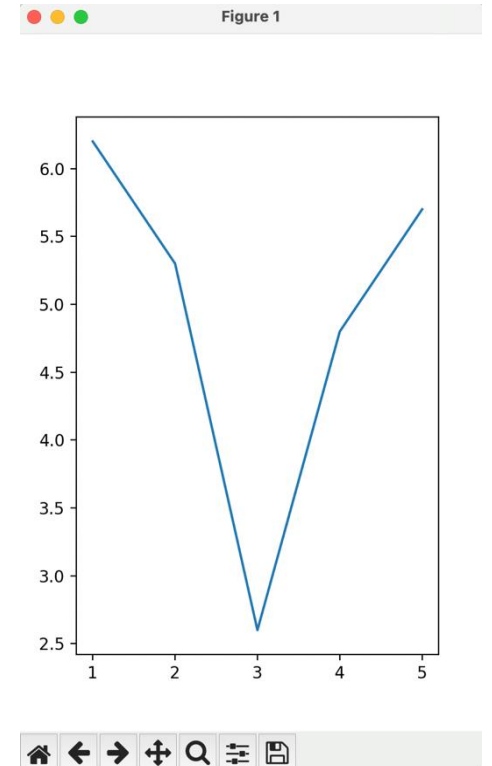
Creating figures

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots(figsize=(4, 6))

a = [1, 2, 3, 4, 5]
b = [6.2, 5.3, 2.6, 4.8, 5.7]

ax.plot(a, b)
plt.show()
```



Creating figures

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots()

a = [1, 2, 3, 4, 5]
b = [6.2, 5.3, 2.6, 4.8, 5.7]

ax.plot(a, b)
```

you can also save figures!

Creating figures

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots()

a = [1, 2, 3, 4, 5]
b = [6.2, 5.3, 2.6, 4.8, 5.7]

ax.plot(a, b)
fig.savefig("important.png")
```

Creating figures

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots()

a = [1, 2, 3, 4, 5]
b = [6.2, 5.3, 2.6, 4.8, 5.7]

ax.plot(a, b)
fig.savefig("important.png")
```

savefig() saves your plot to the given path

Creating figures

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots()

a = [1, 2, 3, 4, 5]
b = [6.2, 5.3, 2.6, 4.8, 5.7]

ax.plot(a, b)
fig.savefig("important.png")
```

it's a method of the **fig** object

Creating figures

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots()

a = [1, 2, 3, 4, 5]
b = [6.2, 5.3, 2.6, 4.8, 5.7]

ax.plot(a, b)
fig.savefig("important.png")
```

you can save the plot as many file types

Creating figures

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots()

a = [1, 2, 3, 4, 5]
b = [6.2, 5.3, 2.6, 4.8, 5.7]

ax.plot(a, b)
fig.savefig("important.jpeg")
```

you can save the plot as many file types

Creating figures

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots()

a = [1, 2, 3, 4, 5]
b = [6.2, 5.3, 2.6, 4.8, 5.7]

ax.plot(a, b)
fig.savefig("important.pdf")
```

you can save the plot as many file types

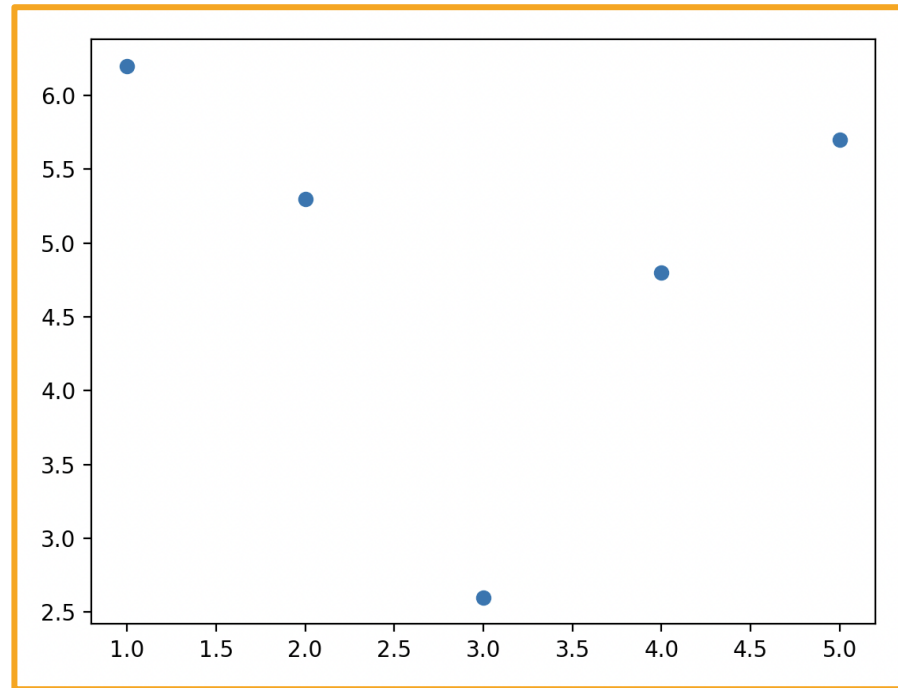
you can make almost any type of
figure in **Matplotlib**!

you can make almost any type of
figure in **Matplotlib**!

some common ones are...

Figure types

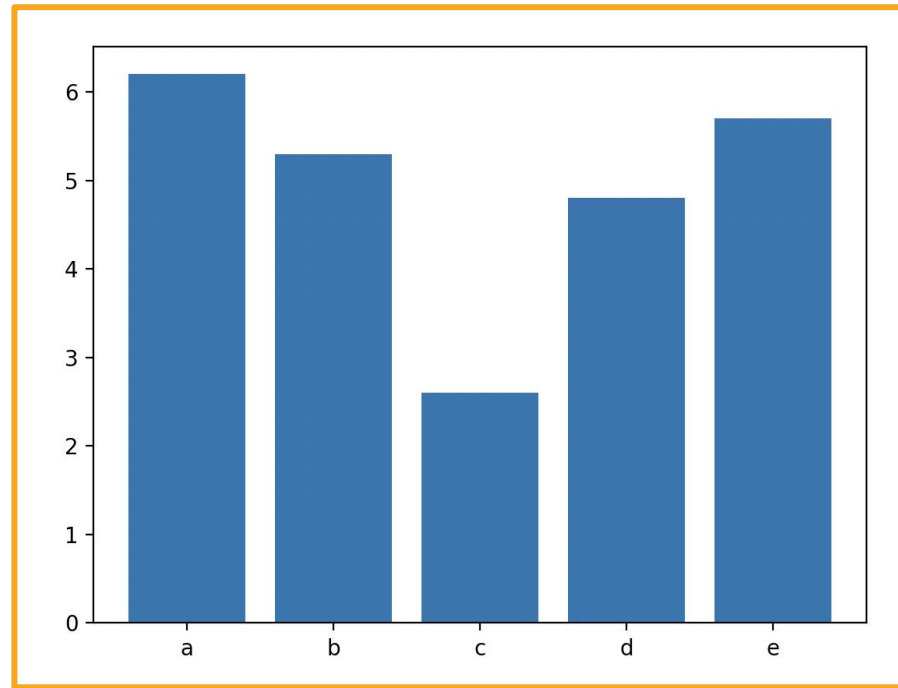
scatterplot



```
ax.scatter(a, b)
```

Figure types

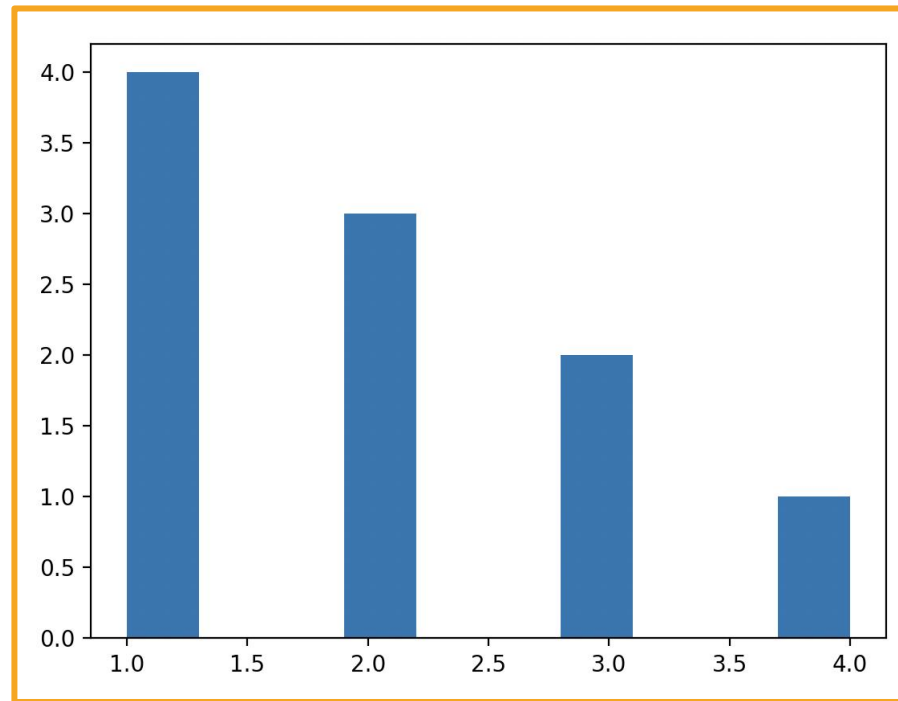
bar chart



```
ax.bar(['a', 'b', 'c', 'd', 'e'], b)
```

Figure types

histogram



```
ax.hist([1, 2, 3, 3, 4, 2, 2, 1, 1, 1])
```

you can also **customize** your figures

Customizing figures

Customizing figures

```
import matplotlib.pyplot as plt
import numpy as np

x, y = np.random.randn(50), np.random.randn(50)

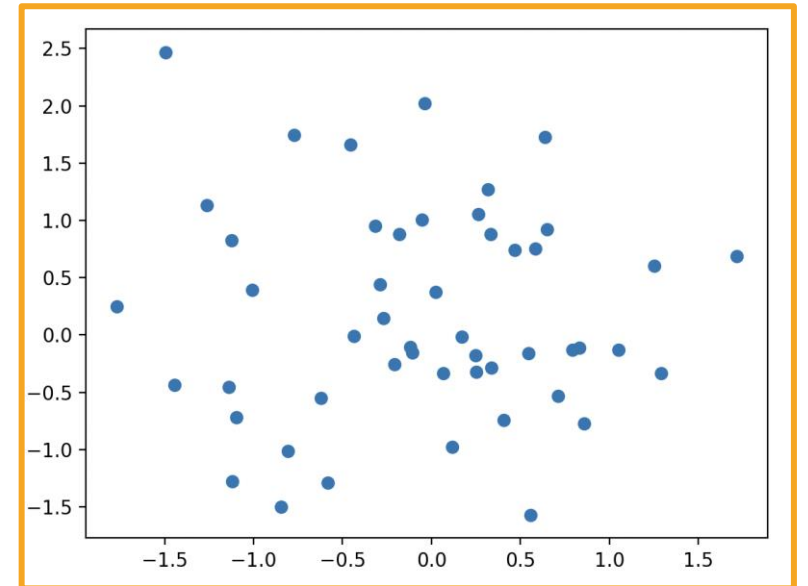
fig, ax = plt.subplots()
ax.scatter(x, y)
plt.show()
```

Customizing figures

```
import matplotlib.pyplot as plt  
import numpy as np
```

```
x, y = np.random.randn(50), np.random.randn(50)
```

```
fig, ax = plt.subplots()  
ax.scatter(x, y)  
plt.show()
```

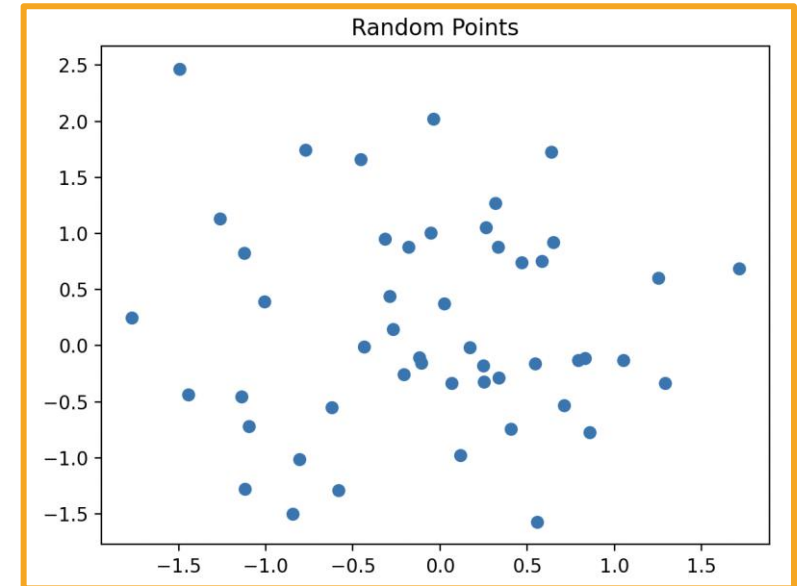


Customizing figures

```
import matplotlib.pyplot as plt
import numpy as np

x, y = np.random.randn(50), np.random.randn(50)

fig, ax = plt.subplots()
ax.scatter(x, y)
ax.set_title("Random Points")
plt.show()
```

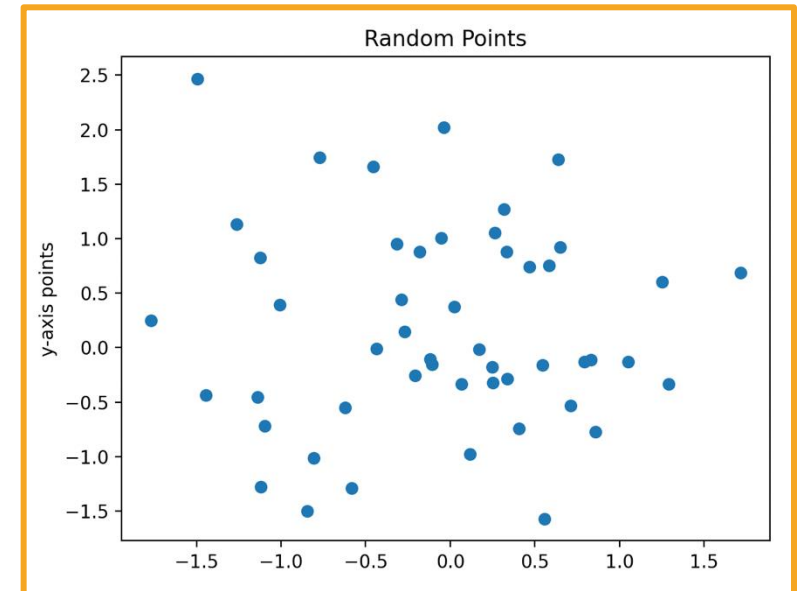


Customizing figures

```
import matplotlib.pyplot as plt
import numpy as np

x, y = np.random.randn(50), np.random.randn(50)

fig, ax = plt.subplots()
ax.scatter(x, y)
ax.set_title("Random Points")
ax.set_ylabel("y-axis points")
plt.show()
```

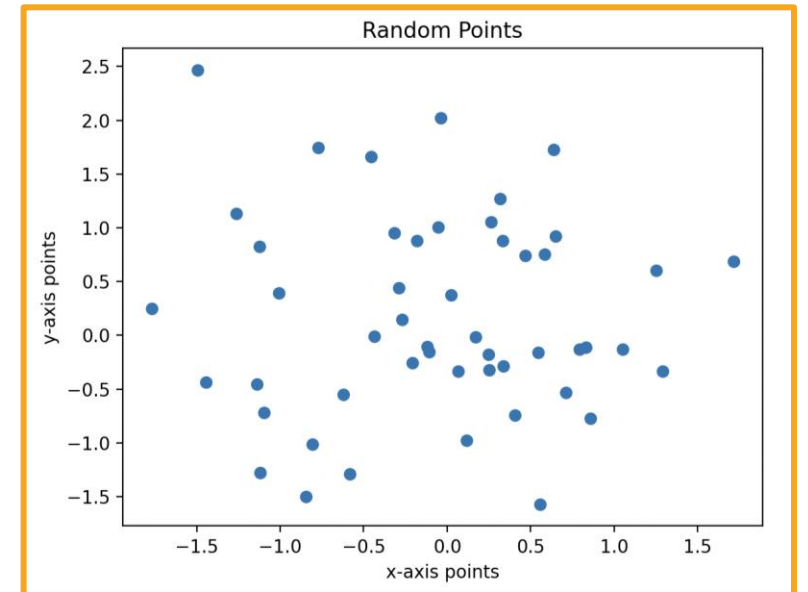


Customizing figures

```
import matplotlib.pyplot as plt
import numpy as np

x, y = np.random.randn(50), np.random.randn(50)

fig, ax = plt.subplots()
ax.scatter(x, y)
ax.set_title("Random Points")
ax.set_ylabel("y-axis points")
ax.set_xlabel("x-axis points")
plt.show()
```



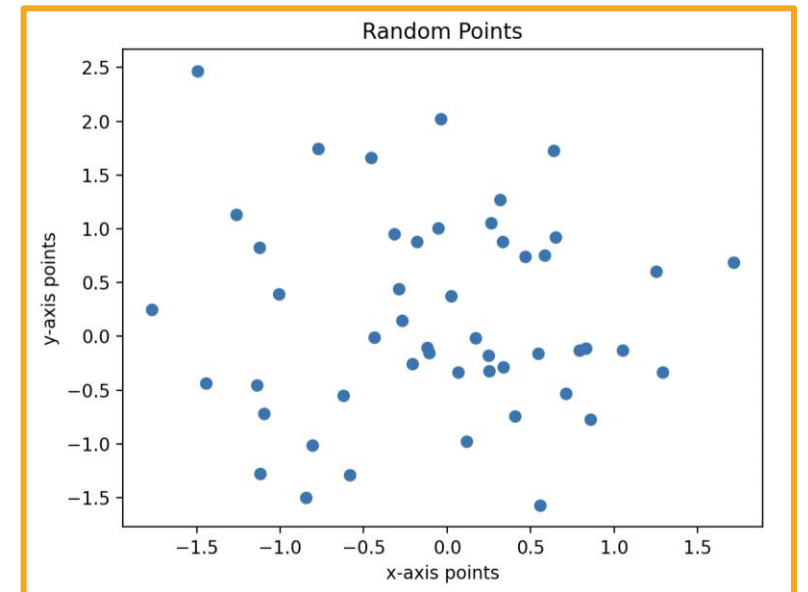
Customizing figures

```
import matplotlib.pyplot as plt
import numpy as np

x, y = np.random.randn(50), np.random.randn(50)

fig, ax = plt.subplots()
ax.scatter(x, y)
ax.set_title("Random Points")
ax.set_ylabel("y-axis points")
ax.set_xlabel("x-axis points")
plt.show()
```

you can further set font, font size, etc...

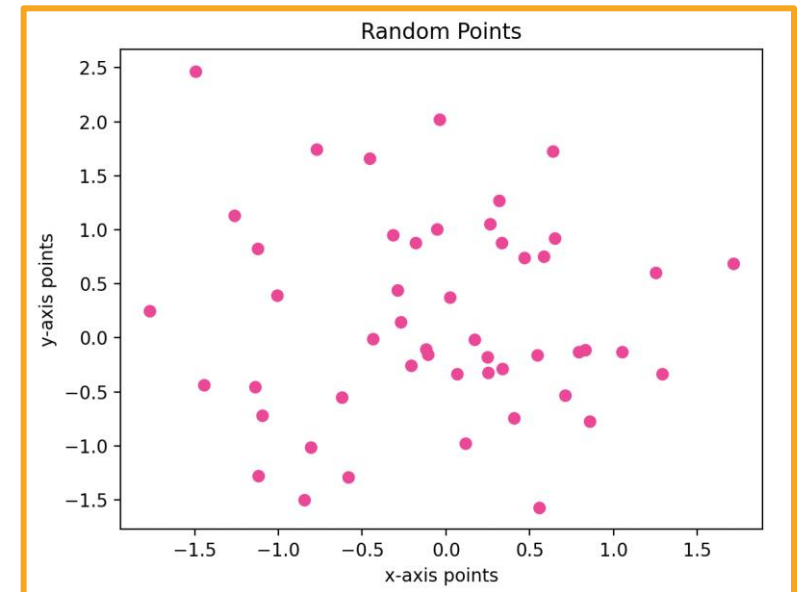


Customizing figures

```
import matplotlib.pyplot as plt
import numpy as np

x, y = np.random.randn(50), np.random.randn(50)

fig, ax = plt.subplots()
ax.scatter(x, y, c="#ff3399")
ax.set_title("Random Points")
ax.set_ylabel("y-axis points")
ax.set_xlabel("x-axis points")
plt.show()
```

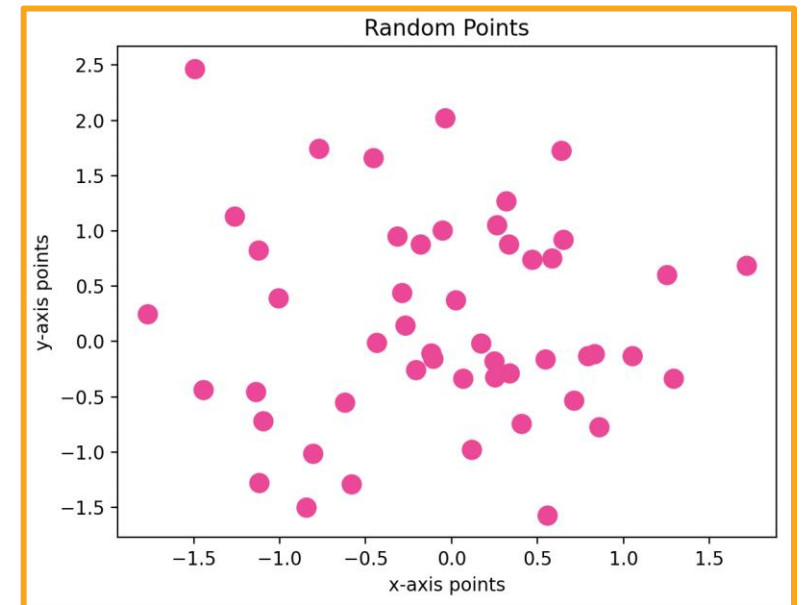


Customizing figures

```
import matplotlib.pyplot as plt
import numpy as np

x, y = np.random.randn(50), np.random.randn(50)

fig, ax = plt.subplots()
ax.scatter(x, y, c="#ff3399",
           s=100)
ax.set_title("Random Points")
ax.set_ylabel("y-axis points")
ax.set_xlabel("x-axis points")
plt.show()
```

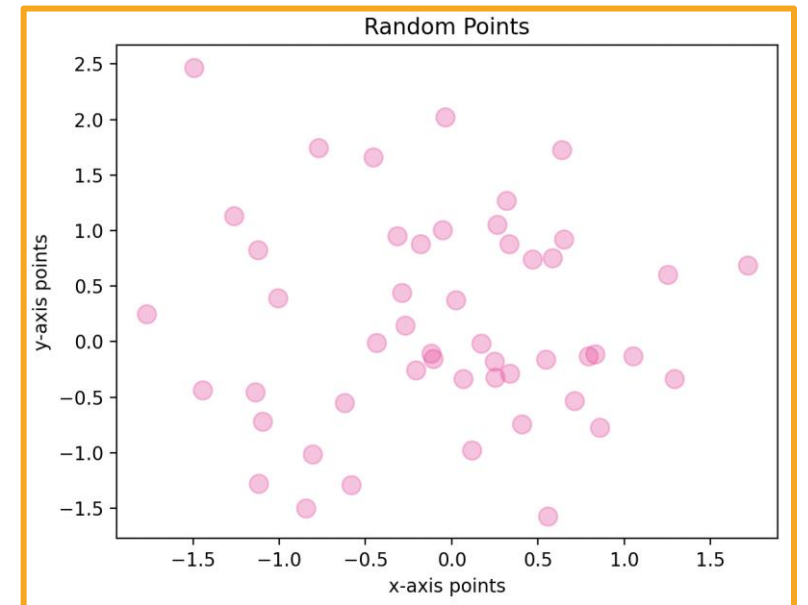


Customizing figures

```
import matplotlib.pyplot as plt
import numpy as np

x, y = np.random.randn(50), np.random.randn(50)

fig, ax = plt.subplots()
ax.scatter(x, y, c="#ff3399",
           s=100, alpha=0.3)
ax.set_title("Random Points")
ax.set_ylabel("y-axis points")
ax.set_xlabel("x-axis points")
plt.show()
```



you can also...

you can also...

set color + size by
an extra variable

you can also...

set color + size by
an extra variable

add legends

you can also...

set color + size by
an extra variable

add legends

make many
subplots within a
figure

you can also...

set color + size by
an extra variable

add legends

make many
subplots within a
figure

make different
kinds of plots

you can also...

set color + size by
an extra variable

add legends

make many
subplots within a
figure

make different
kinds of plots

+ more!

Enter in the chat:

One thing you want to know
how to do with Matplotlib.

Wrapping up...

Wrapping up...

Think about how far you've come!

Wrapping up...

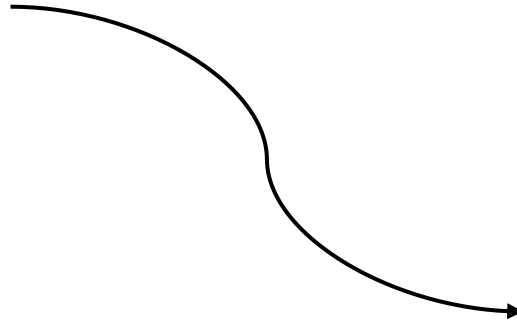
Think about how far you've come!

variables +
statements

Wrapping up...

Think about how far you've come!

variables +
statements



analyzing +
visualizing data

Wrapping up...

Think about how far you've come!

variables +
statements



Be proud of
yourself!

analyzing +
visualizing data

Questions?

(... I'll wait for 2)

For tomorrow:

(07/31)

- Lab 19 *(@ 11:30am)*
- **Final Project!** *(@ 9:00am)*